OBJECTIVE

ZOO202 Animal Diversity Chordates

FINAL TERM PAPER PREPARATION BEST FILE BY

SULMAN ALI

BS-ZOOLOGY 4TH SEMESTER

SPRING 2022

1. Protochord are generally knowan as.....**Lower chordate**
2. Larva of frog is .....**Tadpol**
3. Which mammal have no hair in adult.....**Whales**
4. .....which have cloaca.
5. A coat of hair called.....usually **Pealage**
6. In mammals hair colours depends upon pigment.....**Melanin**
7. Most mammals have..... sets of teeth during lives.**Two**
8. Giraffa and whale have .....neck vertebrae **Cervical**
9. Chiropteria include.....**Bats**
10. Kidney name of mammals. **Metanephric**
12.... Venome Glands are modified.....**Salivary glands**
14. Reptiles tempratures range. **27 to 37**
15. Sperm may be store for up to .....years. **4 years**
16. Penguins belong to..... **Order Sphenisciformes**
17. Pigeon milk produced by.... **Crop**
18.special voice box .....in birds. **Syrinx**
19. birds maintan body temprature at..... **38 to 45°**
20. 1000000/mm.....
21. Reptiles possess.....atria. **Two**
22. Eoalulavis had wing structure called..... **Alula**
23. The beginning of the Tertiary period, about 70 million years ago, was the start of the " age of mammals **Territory period**
24. .....which have altrical. **Robins**
25. Birds excrete..... **Uric acid**
that become dislodged can be rehooked by running a feather through the bill **Hamuli**

1-The allantois is a ventral outgrowth of the gut and stores nitrogenous wastes (e.g., uric acid).

2-bony fishes possess outpockets of the intestine, called

pyloric ceca, which increase absorptive and secretory surfaces

3-The similarities between birds and reptiles are so striking that birds are often referred to as "glorified reptiles."

4-plate of bone, called the secondary palate, evolved in the archosaurs and separates the nasal and mouth passage ways.

5.Tail loss, or autotomy, is an adaptation that allows a lizard to escape from a predator's grasp, or the disconnected, wiggling piece of tail may distract a predator from the lizard.

6-The tongues of turtles and crocodilians are non-protrusible and aid in swallowing.

7-Most reptiles excrete uric acid. It is nontoxic, and being relatively insoluble in water, it precipitates in the excretory system.

8-Members of the genus Lampetra are called brook lampreys.

9-Eggs hatch in approximately three weeks into ammocoete larvae.

10-a group of fishes called hag-fishes are the most primitive vertebrates known (living or extinct).

11-It was present in the bony armor of a group of fishes called ostracoderms.

12-In addition to their biting mouthparts and paired appendages, chondrichthians possess epidermal placoid scales and a cartilaginous endoskeleton.

13-A third group of sarcopterygians, called osteolepiforms.

14- Long gill processes, called gill rakers, trap plankton while the fish is swimming through the water with its mouth open.

15- Reptiles in the subclass Anapsida lack openings or fenestrae region of the skull.

16- The approximately 225 species of turtles are characterized by a bony shell, limbs, articulating internally to the ribs, and a keratinized beak rather than teeth.

17- The dorsal portion of the shell is the carapace.

18- The ventral portion of the shell is the plastron.

19- Turtles have eight cervical vertebrae that can be articulated into an S-shaped.

20- All turtles are oviparous they lay and cover with soil clutches of 5 to 100 eggs.

21- About 3,300 species of lizards are in the suborder Sauria.

22- Lizards vary in length from only a few centimeters to as large as 3 m.

23- Geckos, commonly

found on the walls of human dwellings in semitropical area.

24- Iguanas have robust bodies, short necks, and distinct heads.

25- Chameleons are adapted to arboreal lifestyles and use a long, sticky tongue to capture insects.

26- About 2,300 species are in the suborder Serpentes.

27- The skin of amphisbaenians has ringlike folds called annuli.

28- The order Crocodilia has 21 species.

29- Crocodilians have not changed much over their 170 million-year history.

30- The crocodilians stomach is gizzardlike, and crocodilians swallow rocks and other objects as abrasives for breaking apart ingested food.

31-Reptilian skin is thick, dry, and keratinized

32- All reptiles periodically shed the outer, epidermal layers of the skin in a process

called ecdysis

33- Tail loss, or autotomy, is an adaptation that allows a lizard to escape from a predator's grasp, or the disconnected, wiggling piece of tail may distract a predator from the lizard.

34- Skull and jaws of reptiles help to capture large prey.

35- Venom glands are modified salivary glands.

36 In most reptiles, a negative-pressure mechanism is responsible for lung ventilation.

37- To sustain activity, however, body temperatures are regulated within a narrow range, between 25 and 37° C.

38- Heat loss from these groups, called hibernacula,.

39- In snakes and some lizards, the upper and lower eyelids fuse in the embryo to form a protective window of clear skin, called the spectacle.

40- A kidney with many more blood-filtering units, called nephrons

41- The functional kidneys of adult reptiles are called metanephric kidneys.

42- Lizards and snakes possess paired hemipenes at the base of the tail that are erected by being turned inside out, like the finger of a glove.

43- Gonads lie in the abdominal cavity.

44- Birds have horny beaks and scaly feet.

❖ 45- Birds' complex mating systems and behavioral patterns increase the chances of offspring survival

46- In 1861, one of the most important vertebrate fossils was found in a slate quarry in Bavaria, Germany.

47- Sinornis fossils are 135 million years old—only 15 million years younger than Archaeopteryx.

48- Eoalulavis, was found in 115- million-year-old deposits and had a wingspan of 17 cm (roughly the same as a goldfinch).

49- . It is sufficient to say that adaptive radiation has resulted in about 9,100 species of living birds, which are divided into about 27 orders (table 21.1).

50- The covering of feathers on a bird is called the plumage.

51- Preening, which is done by rubbing the bill over the feathers, keeps the feathers smooth, clean, and in place.

52- Birds also have a reduced number of skull bones, and a lighter, keratinized sheath called a bill replaces the teeth.

53- The posterior caudal vertebrae are fused into a pygostyle, which helps support the tail feathers that are important in steering.

54- The sternum of most birds bears a large, median keel for the attachment of flight muscles.

55- Flapping flight generates the power for flight and is the most common type of flying.

56- Hummingbirds perform hovering flight.

57- The crop of pigeons produces "pigeon's milk

58- The bird heart is relatively large (up to 2.4% of total body weight), and it beats rapidly. Rates in excess of 1,000 beats per minute have been recorded for hummingbirds under stress.

59- The heart rate of an ostrich, for example, varies between 38 and 176 beats per minute.

60- • A special voice box, called the syrinx, The lungs of birds are made of small

air tubes called parabronchi.

61- Himalayas at altitudes of 9,200 m. A human mountain climber begins to feel the symptoms of altitude sickness and must carry auxiliary oxygen supplies above 2,500 m.

Topic no 62

62 Birds maintain body temperatures between 38 and 45° C

63- Lethal extremes are lower than 32 and higher than 47° C

64- Shivering also generates heat in extreme cold

65-whippoorwills allow their body temperatures to drop from about 40° C to near 16° C, and respiratory rates become very slow.

66-The corpus striatum functions in visual learning, feeding, courtship, and nesting.

67-• A pineal body on the roof of the forebrain appears to stimulate ovarian development and to regulate other functions influenced by light and dark periods.

68-The optic tectum (the roof of the mid brain), along with the corpus striatum, plays an important role in integrating sensory functions.

69-Vision is an important sense for most birds.

70-Cones are especially concentrated (1,000,000/mm2) at a focal point called the fovea.

71-The one at the center of the retina is sometimes called the The one at the center of the retina is sometimes called the "search fovea" .

72-The posterior fovea is called the "pursuit fovea" because binocular vision produces depth perception, which is necessary to capture prey.

73-Pigeons have eyes located well back on the sides of their head, giving them a nearly

360° monocular field, but a narrow binocular field.

74-Loose, delicate feathers called auriculars cover the external ear opening.

75-The sensitivity of the avian ear (100 to 15,000 Hz) is similar to that of the human ear (16 to 20,000 Hz).

76-Like reptiles, birds excrete uric acid, which is temporarily stored in the cloaca

77-some birds have supraorbital salt glands that drain excess sodium chloride through the nasal openings to the outside of the body.

78-All birds are oviparous.

79- Gonads are in the dorsal abdominal region, next to the kidneys.

80-In females, two ovaries form during development, but usually only the left ovary fully develops .

81-Except for certain waterfowl and ostriches, birds have no intromittent organ.

82-The oviduct opens into the cloaca.

83-Most birds are monogamous

84-Prairie chickens are polygynous, and males display in groups called leks

85-The group of eggs laid and chicks produced by a female is called a clutch.

86-The incubation period lasts between 10 and 80 days.

87-The Bird 🐓that are dependent on their parents are said to be altricial.

88- American robins weigh 4 to 6 g at hatching and leave the nest 13 days later weighing 57 g.

89-Some long distance migrants may store fat equal to 50% of their body weight and make nonstop journeys.

90-• In clear weather, many birds fly at altitudes greater than 1,000 m, which reduces the likelihood of hitting tall obstacles.

91-Birds' lenses are transparent to ultraviolet light, and their photoreceptors respond to it, allowing them to orient using the sun, even on cloudy days. This orientation cue is called a sun compass.

92- The angle between the North Star and the horizon decreases as you move toward the equator.

93-Any member of the group of vertebrate animals in which the young are nourished

with milk from special mammary glands of the mother.

94-Hair is a typical mammalian feature, although in

many whales it has disappeared except in the fetal stage.

A chain of three tiny bones transmits sound waves across the middle ear.

A muscular diaphragm separates the heart and the lungs from the abdominal cavity.

97-Mature red blood cells (erythrocytes) in all mammals lack a nucleus; all other vertebrates have nucleated red blood cells.

98-Mammalian characteristics evolved gradually over a

200-million-year period in the synapsid lineage.

99-Two subclasses of mammals evolved during the Mesozoic era—the Prototheria and the Theria.

100-The beginning of the Tertiary period, about 70 million years ago, was the start of the “age of mammals.

101-By the middle of the Permian period , other successful mammal-like reptiles had arisen from the pelycosaurs. They were a diverse group known as the therapsids.

102-• The last therapsids were a group called the cynodonts.

103-• The first mammals were small (less than 10 cm long) with delicate skeletons.

104-The infraclass Ornithodelphia contains the monotremes.

105-The six species of monotremes are found in

Australia and New Guinea.

106-About 250 species of marsupials live in the Australian region and the Americas.

107-The approximately 3,800 species of eutherians are classified into 17 orders.

108-Mammals are naturally distributed on all continents

except Antarctica, and they live in all oceans.

109-Hair is seated in an invagination of the epidermis, called a hair follicle.

110-A coat of hair, called pelage.

111- Guard hairs may sometimes be modified into thick-shafted hairs called vibrissae.

112-A band of smooth muscle, called the arrector pili muscle.

113-Arrector pili muscles are under the control of the

autonomic nervous system, which also controls a mammal’s “fight-or-flight” response.

114-Pelage is reduced in large mammals from hot climates.

115-Glands develop from the epidermis of the skin.

116-Sebaceous (oil) glands are associated with hair follicles, and their oily secretion lubricates and waterproofs the skin and hair.

117-Most mammals also possess sudoriferous (sweat) glands.

118-Small sudoriferous glands (eccrine glands) release watery secretions used in evaporative cooling.

119-Scent or musk glands are around the face, feet, or anus of many mammals.

120-Larger sudoriferous glands (apocrine glands) secrete a mixture of salt, urea, and water, which microorganisms on the skin convert to odorous products.

121-Mammary glands are functional in female mammals and are present, but nonfunctional, in males.

122-• One feature that zoologists use to distinguish reptilian from mammalian skulls is the method of jaw articulation.

123-• In mammals, the secondary palate extends posteriorly by a fold of skin, called the soft palate.

124-• In reptiles, the teeth are uniformly conical, a condition referred to as homodont.

125-The first teeth emerge before or shortly after birth and are called deciduous or milk teeth.

126-In mammals, the teeth are often specialized for different functions, a condition called heterodont.

126-In rodents, the incisors grow throughout life.

127-The vertebral column of mammals is divided into five regions.

128-• A cecum is a fermentation pouch where microorganisms aid in cellulose digestion.

129-Sheep, cattle, and deer are called ruminants (L. ruminare, to chew the cud).

130-The evolution of similar structures in different lineages is called convergent evolution.

131-Expiration is normally by elastic recoil of the lungs and relaxation of inspiratory muscles, which decreases the volume of the thoracic cavity.

132-Shivering thermogenesis is muscular activity that generates large amounts of heat but little movement.

133-walrus in cold, arctic waters has a surface temperature near 0° C; however, a fewcentimeters below the skin surface, body temperatures are about 35° C.

134- most of the body of an arctic mammal is unusually well insulated.

135-Countercurrent heat-exchange systems may help regulate heat loss from exposed areas .

136-Hibernation is a period of winter inactivity in which the hypothalamus of the brain slows the metabolic, heart, and respiratory rates.

137-Rodentia (e.g.,chipmunks and woodchucks), and Chiroptera (bats).

138-After a hibernating mammal retreats to a burrow or a nest, the hypothalamus sets the body's thermostat to about 2° C.

139-The respiratory rate of a hibernating ground squirrel falls from 100 to 200 breaths per minute to about four breaths per minute.

140-The heart rate falls from 200 to 300 beats per minute to about 20 beats per minute.

141-During hibernation, a mammal may lose a third to half of its body weight.

142-The sensory patch of the inner ear that contains the sound receptors is long and coiled and is called the cochlea.

143-Color vision is less well developed in mammals than in reptiles and birds.

144-The primary adaptation of the mammalian nephron is a portion of the tubule system called the loop of the nephron.

145-beavers produce urine that is twice as concentrated as blood, while

Australian hopping mice produce urine that is 22 times more concentrated than Blood.

146-In the nephron of the kidney, fluids and small solutes are filtered from the blood through the walls of a group of capillary like vessels, called the glomerulus.

147-The primary adaptation of the mammalian nephron is a portion of the tubule system called the loop of the nephron.

148-Metabolic oxidation of carbohydrates produces water as a byproduct.

149-Visual cues are often used in communication.

150-Vocalizations and tactile communication are important in primate social interactions.

151-Development is arrested for the three months during which the recently born young do most of their nursing. This mechanism is called embryonic ddiapause

152-In viviparous mammals, the oviducts are modified into one or two uteri

153-Many mammals are monestrus and have only a single yearly estrus cycle that is sharply seasonal.

154-Rats and mice have estrus cycles that repeat every

four to six days.

155- In mammals Females store sperm in the uterus for periods in excess of two months.

156-Female echidnas incubate eggs in a ventral pouch. 157-Platypus eggs are laid in their burrows.

158-The marsupial gestation period (the length of time young develop within the female reproductive tract) varies between ▪ 8 and 40 days in different species.

159-After birth, tiny young crawl into the marsupium and attach to a nipple, where they suckle for an additional 60 to 270 days

160-Gestation periods of eutherian mammals vary from

20 days (some rodents) to 19 months (the African elephant)